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1 %creates Bode plot for rotating electric toothbrush
2 syms s J b L R KT Kb
3
4 G = KT / (J*L*s^2 + (J*R + b*L)*s + (b*R + KT*Kb));
5
6 %estimated system parameters
7 Jv = 6.05e-7; %kgm^2
8 bv = 3.78e-7 ; %Nms/rad
9 Lv = 1.0e-3 ; %henry
10 Rv = 5; %ohms
11 KTv = 2e-3; %Nm/A
12 Kbv = 2e-3;%Nm/A
13
14 % values to induce resonance
15 % Jv = 6.05e-7;
16 % bv = 0;
17 % Lv = 1e-3;
18 % Rv = 0.01;
19 % KTv = 2e-3;
20 % Kbv = 2e-3;
21 G_num = subs(G, [J b L R KT Kb], [Jv bv Lv Rv KTv Kbv]);
22
23 [num, den] = numden(G_num);
24 num = sym2poly(num);
25 den = sym2poly(den);
26
27 sys = tf(num, den);
28 h = bodeplot(sys);
29 set(findall(h.Children, 'Type', 'Line'), 'LineWidth', 2)
30 grid on
31
32 % calculate zeta
33 zetanum = Rv*Jv + bv*Lv;
34 zetadenom = 2*sqrt(Jv*Lv*(bv*Rv+KTv*Kbv));
35 zeta = zetanum/zetadenom
36 natfreqnum = bv*Rv+KTv*Kbv;
37 natfreqdenom = Jv*Lv;
38 natfreq = sqrt(natfreqnum/natfreqdenom)

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